Xujiang Tang

Online Learning, Bandit Algorithms, and Mathematical Machine Learning

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RESEARCH PROFILE

Undergraduate researcher preparing for doctoral study in online learning, multi-armed bandits, reinforcement learning, and sequential decision-making. Strong mathematical preparation in probability, statistics, and optimization, with experience translating theoretical ideas into reproducible Python and PyTorch implementations.

EDUCATION

Visiting Student 2025 - Present

The Hong Kong University of Science and Technology (Guangzhou) | Advisor: Prof. Tianyuan Jin

- Research preparation in theoretical machine learning, online learning, bandit algorithms, reinforcement learning, and stochastic optimization.
- Systematic study of concentration, martingales, KL divergence, exponential families, and information-theoretic lower bounds for adaptive decisions.

B.Sc. in Mathematics and Applied Mathematics Sep 2023 - Jun 2027

Yangtze University College of Arts and Sciences, Jingzhou, China | GPA: 3.5/4.0

- Optimization Theory: 97/100, ranked first in the major; Numerical Computation Methods: 96/100.
- Relevant coursework: probability and mathematical statistics, optimization, numerical computation, mathematical analysis, advanced algebra, differential equations, mathematical modeling, and programming.

RELEVANT RESEARCH PREPARATION

Online Learning, Bandits, and Sequential Decisions 2025 - Present

Research Training, HKUST(GZ) | Advisor: Prof. Tianyuan Jin

- Studied regret minimization, best-arm identification, exploration-exploitation tradeoffs, and adaptive decision-making.
- Authored eight LaTeX notes connecting adaptive probability to confidence bounds, exponential-family models, and change-of-measure lower bounds.
- Developed a focused research agenda around sample- and resource-efficient sequential learning.

Bilevel Optimization and Implicit Differentiation 2025 - Present

Independent Research

- Investigated hypergradient computation beyond invertible-Hessian assumptions using differentiable lower-level solution manifolds.
- Developed a curvature-filter framework based on projection operators and singular behavior near focal sets; submitted to *TMLR*.

Peking University Open-Source LLM Group Jun 2024 - Aug 2024

Research and Engineering Intern

- Contributed to large-language-model deployment and retrieval-augmented generation systems.
- Implemented retrieval and knowledge-integration modules for external knowledge use.

ADDITIONAL RESEARCH EXPERIENCE

Deep Hedging under Rough Volatility 2025 - Present

Independent Research

- Developed a fractional-kernel neural architecture for variance-optimal hedging under rough-volatility models.
- Integrated stochastic analysis, neural approximation, and numerical optimization.

Dynamical Systems and Koopman Operator Learning Mar 2024 - Present

Research Collaborator, Prof. Yanbing Jia's Group

- Proposed a Koopman Operator Linearization Skeleton for long-horizon nonlinear prediction.
- Implemented and evaluated the framework on multiple chaotic-system benchmarks against deep-learning baselines.

PUBLISHED AND ACCEPTED

1. Q. Li and **X. Tang**. HKGEduRec: A Knowledge Graph-Enhanced Dynamic Hybrid Framework for Educational Recommendation with Cold-Start Mitigation. *IEEE ICMEIM 2025*, pp. 1-5. EI-indexed.
2. **X. Tang** and Q. Li. Logical Gene Encoding: A Bio-Inspired Approach for Energy-Efficient Automated Reasoning. *IEIT 2025*. EI-indexed. DOI: 10.2991/978-94-6463-803-5_81.
3. Q. Li and **X. Tang**. Robust Optimal Reinsurance and Investment with Inflation Risk: A Game-Theoretic Approach and Explicit Solutions. *AIMS Mathematics*. Accepted.

UNDER REVIEW AND SUBMITTED

1. **X. Tang et al.** The Curvature Filter in Bilevel Optimization: Implicit Differentiation Without Nondegeneracy. *Transactions on Machine Learning Research*. Submitted.
2. Y. Fan, **X. Tang**, and R. Guan. Deep Hedging under Rough Volatility: A Fractional Kernel Embedding Approach with Optimal Convergence Rate. *AIMS Mathematics*. Under review.
3. **X. Tang et al.** Structure-preserving Koopman Predictive Control for Memristive Neural Dynamics: Input-exact and Commutator-defect Lifting. *Biological Cybernetics*. Under review.
4. **X. Tang et al.** Spectral Network Determinants of Seizure-like Synchronization and Spread in Coupled Hindmarsh-Rose Brain Models. *Cognitive Neurodynamics*. Under review.

SELECTED TECHNICAL NOTES

Eight reproducible LaTeX notes on sequential decisions: regret and exploration; adaptive data and filtrations; concentration bounds; martingales and optional stopping; KL divergence; exponential-family bandit models; and change-of-measure lower bounds. Available at txj2006.github.io/notes/.

HONORS

Kaggle Silver Medal, Jigsaw Toxic Comment Classification Challenge | Regional Gold Medal, WorldQuant International Quant Championship | Third Prize, UCAS Graduate AI Forum | National Second Prize, National College Students Statistical Modeling Competition | National Second Prize, Hua Zhong Cup Mathematical Modeling Competition

SKILLS

Programming and tools: Python, PyTorch, NumPy, Pandas, C++, R, MATLAB, Git/GitHub, Linux, LaTeX.
Theory and methods: online learning, bandit algorithms, probability, statistical modeling, optimization, stochastic simulation, LLM deployment, retrieval-augmented generation.
Languages: Mandarin Chinese; English.